Computer Vision: Applications

Object Detection · IoU · NMS · mAP · FPN · Anchor-Free Methods · Focal Loss

Intersection over Union (IoU)

What is IoU?

-IoU: The standard metric for measuring bounding box overlap quality.

- Formula: IoU = Area of Intersection / Area of Union
- Range: 0.0 (no overlap) to 1.0 (perfect overlap).
- Threshold-based decision: IoU > 0.5 = correct detection (PASCAL VOC standard).
- COCO benchmark uses IoU > 0.75 for stricter evaluation.
- Used in: RPN anchor labeling, NMS, mAP computation, and regression losses.
- A prediction with IoU < threshold is counted as a False Positive (FP).

IoU Variants

Standard IoU

- Area(A ∩ B) / Area(A ∪ B)
- Does not penalize non-overlapping boxes
- Gradient vanishes when boxes don't overlap
- Used mainly for evaluation metrics

Advanced IoU Losses

- GIoU: adds penalty for enclosing area
- DIoU: adds center-point distance penalty
- CIoU: adds aspect ratio consistency term
- Used in YOLO v4 / v5 / v8 regression loss

Non-Maximum Suppression (NMS)

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-NMS: Eliminates duplicate boxes, keeping only the best detection per object.

- Problem: detectors produce many overlapping boxes for the same object.
- Step 1: Sort all predicted boxes by confidence score (highest first).
- Step 2: Keep the top-scoring box.
- Step 3: Remove all boxes with IoU > threshold (e.g. 0.5) with the kept box.
- Step 4: Repeat for remaining boxes until none are left.
- Soft-NMS: reduces score of overlapping boxes instead of deleting — better for dense scenes.
- WBF (Weighted Boxes Fusion): merges overlapping boxes — used in model ensembles.

Mean Average Precision (mAP)

Precision, Recall & mAP

-mAP: The primary benchmark metric for object detection models.

- Precision = TP / (TP + FP) — of all predicted boxes, how many were correct?
- Recall = TP / (TP + FN) — of all ground-truth objects, how many were found?
- AP (Average Precision) = area under the Precision-Recall curve for ONE class.
- mAP = mean of AP across ALL object classes.
- mAP@0.5: computed at IoU threshold 0.5 (PASCAL VOC standard).
- mAP@[0.5:0.95]: averaged over IoU thresholds 0.50 to 0.95, step 0.05 (COCO).
- Higher mAP = better detector. YOLOv8-L achieves ~53 mAP on COCO.

Feature Pyramid Network (FPN)

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-FPN: Solves multi-scale detection using the CNN's own feature hierarchy.

- Problem: a single feature map cannot detect both tiny and large objects well.
- Bottom-up pathway: standard CNN forward pass generating feature maps at different scales.
- Top-down pathway: upsample semantic-rich features from deeper layers.
- Lateral connections: 1×1 conv matches channel dimensions before element-wise addition.
- Outputs P2–P5: each scale detects objects of a specific size range.
- P2 = small objects (high resolution), P5 = large objects (low resolution).
- Used in: Faster R-CNN+FPN, RetinaNet, Mask R-CNN, YOLO v8.

Anchor-Free Detection Methods

Anchor-Based vs Anchor-Free

Anchor-Based (YOLO v2-v5, Faster R-CNN)

- Uses predefined anchor boxes per location
- Requires careful anchor size/ratio tuning
- IoU matching to assign GT to anchors
- 9–15 anchors per feature map location
- Sensitive to anchor size mismatches

Anchor-Free (FCOS, YOLO v8, DETR)

- Predicts box directly from center point
- No anchor design required
- Centerness branch reduces false positives
- Simpler, cleaner training pipeline
- DETR uses transformer attention for detection

RetinaNet & Focal Loss

Focal Loss — Solving Class Imbalance

-RetinaNet: First one-stage detector to match two-stage accuracy using Focal Loss.

- Problem: ~100,000 anchors per image but only 10–100 contain objects.
- Background anchors dominate training — overwhelming the gradient signal.
- Standard cross-entropy gives equal weight to easy and hard examples.
- Focal Loss formula: $FL(p) = -\alpha (1 - p)^\gamma \cdot \log(p)$
- γ (gamma) = focusing parameter. $\gamma=0 \rightarrow$ standard CE. $\gamma=2 \rightarrow$ best in practice.
- α = class balancing weight to handle foreground/background imbalance.
- Effect: down-weights easy negatives, focuses training on hard misclassified examples.

Numerical 1

N1: Box A = [2, 3, 6, 7] and Box B = [4, 5, 8, 9] (corner format x1,y1,x2,y2). Calculate IoU.

Solution:

1. Intersection: $x1=\max(2,4)=4$, $y1=\max(3,5)=5$, $x2=\min(6,8)=6$, $y2=\min(7,9)=7$
2. Intersection area = $(6-4) \times (7-5) = 2 \times 2 = 4$

Answer: IoU ≈ 0.14 — weak overlap, rejected at IoU threshold 0.5 (False Positive)

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Solution:

$$\text{Area of A} = (6-2) \times (7-3) = 4 \times 4 = 16$$

$$\text{Area of B} = (8-4) \times (9-5) = 4 \times 4 = 16$$

$$\text{Union} = 16 + 16 - 4 = 28$$

$$\text{IoU} = 4 / 28 \approx 0.143$$

Answer: IoU ≈ 0.14 — weak overlap, rejected at IoU threshold 0.5 (False Positive)